Functions ~ the main functions of any door are to:

1. Provide a means of access and egress.
2. Maintain continuity of wall function when closed.
3. Provide a degree of privacy and security.

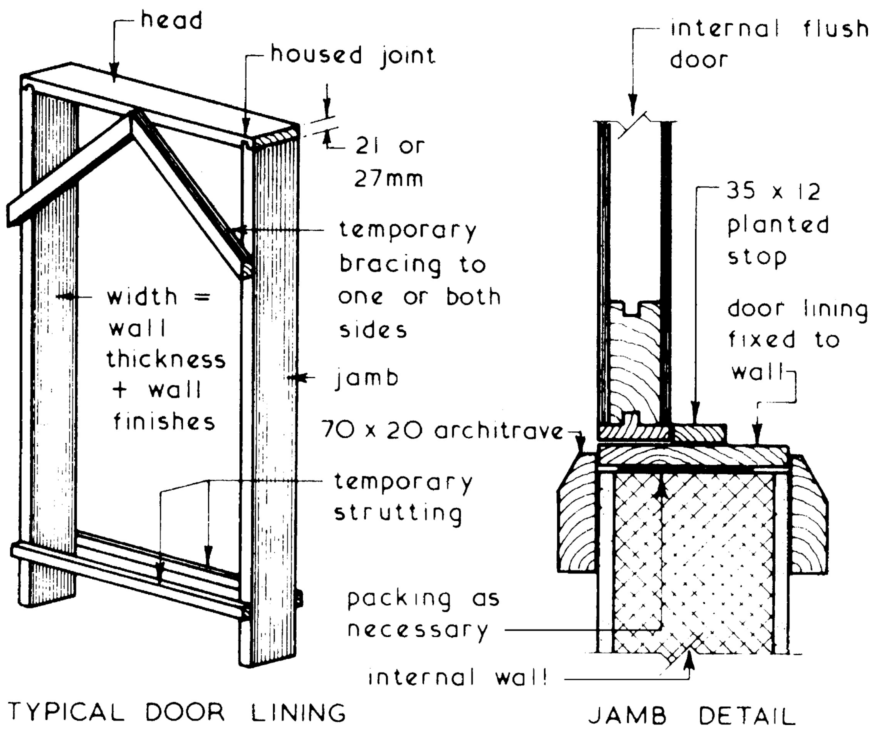
Choice of door type can be determined by:

1. Position – whether internal or external.
2. Properties required – fire resistant, glazed to provide for borrowed light or vision through, etc.
3. Appearance – flush or panelled, painted or polished, etc.

Door Schedules ~ these can be prepared in the same manner and for the same purpose as that given for windows on page 460.

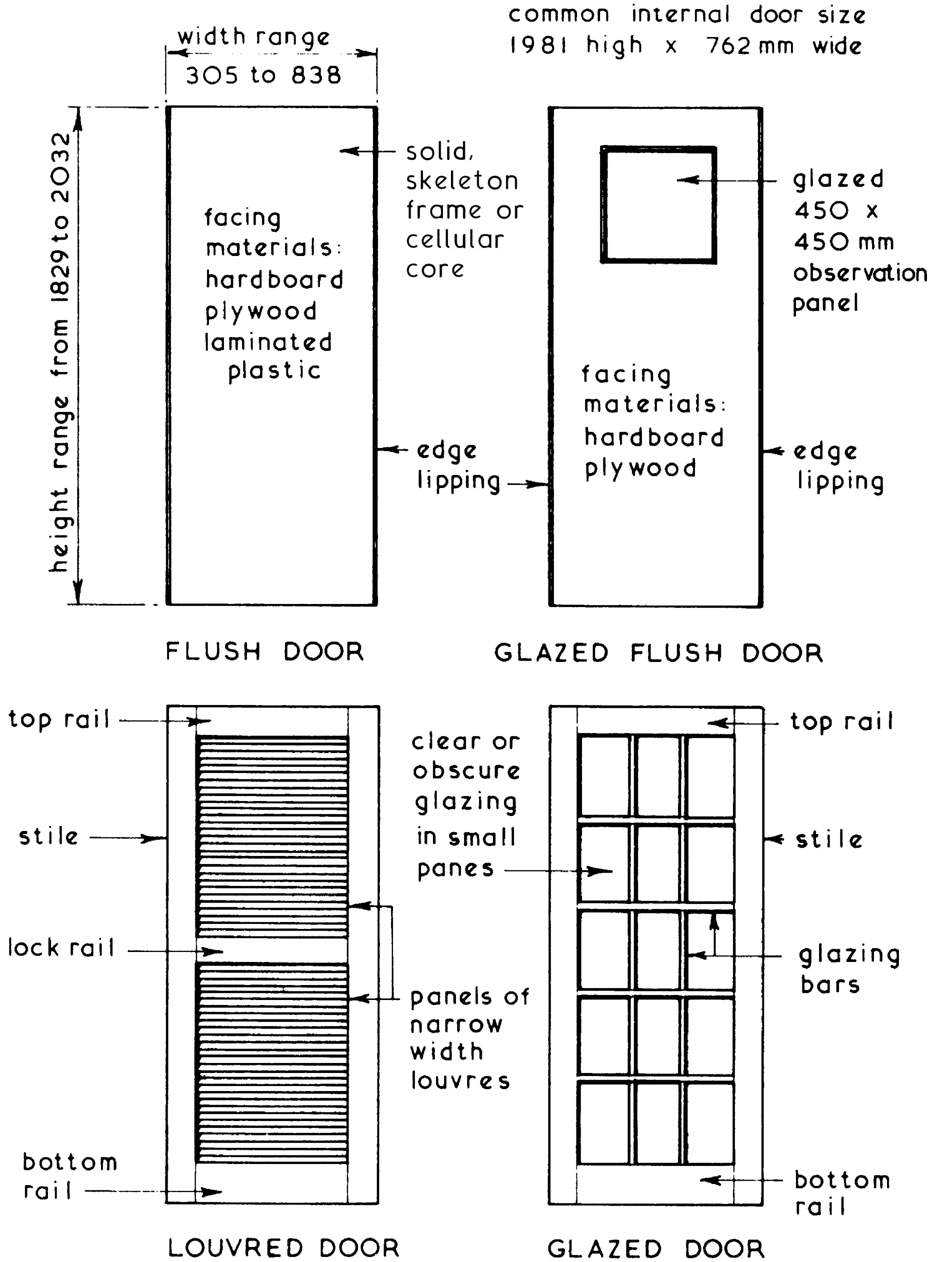
Internal Doors ~ these are usually lightweight and can be fixed to a lining, if heavy doors are specified these can be hung to frames in a similar manner to external doors. An alternative method is to use doorsets which are usually storey height and supplied with prehung doors.

Typical door Lining Details ~



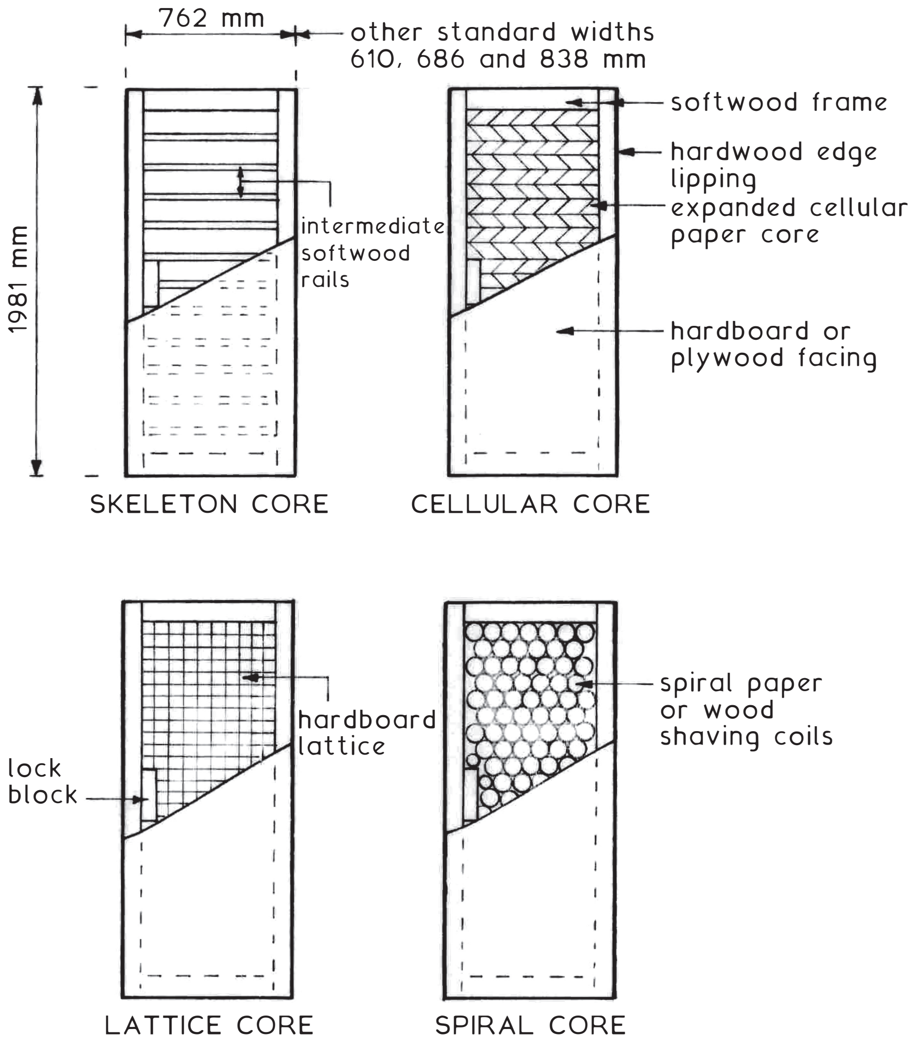
Internal Doors ~ these are similar in construction to the external doors but are usually thinner and therefore lighter in weight.

Typical Examples ~



Flush doors of lightweight construction are suitable to access many interior situations. If there is an additional requirement for resistance to fire, the construction will include supplementary lining as shown on pages 883 and 884. Fire resisting and non-fire resisting doors can be produced with identical flush finishes, but the fire doors must be purposely labelled as shown on page 882.

Examples of Non-fire Doors for Interior Use ~

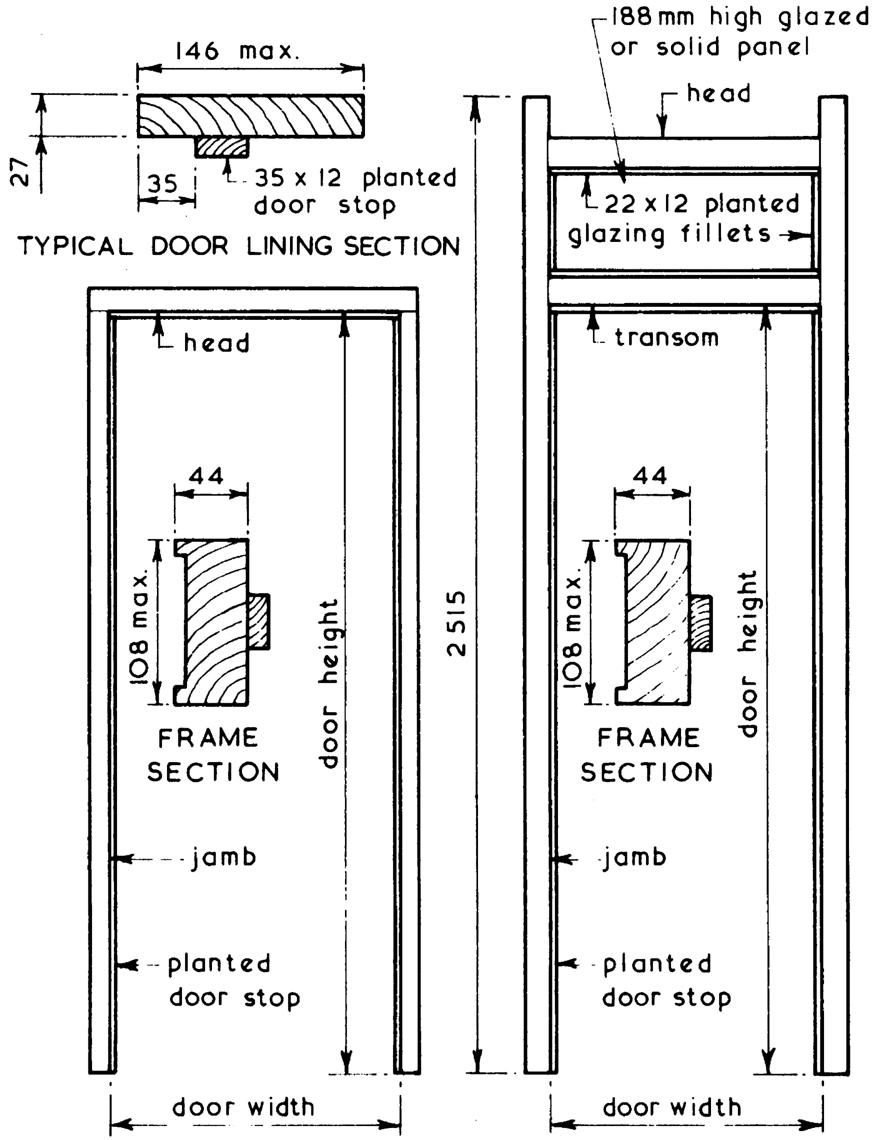


Note 1: Overall thickness can vary between manufacturers. Typically 35 or 40 mm.

Note 2: Height and width dimension of 2032 mm × 813 mm also available from most joinery producers.

Internal Door Frames and Linings ~ these are similar in construction to external door frames but usually have planted door stops and do not have a sill. The frames are sized to be built in conjunction with various partition thicknesses and surface finishes. Linings with planted stops are usually employed for lightweight domestic doors.

Typical Examples ~



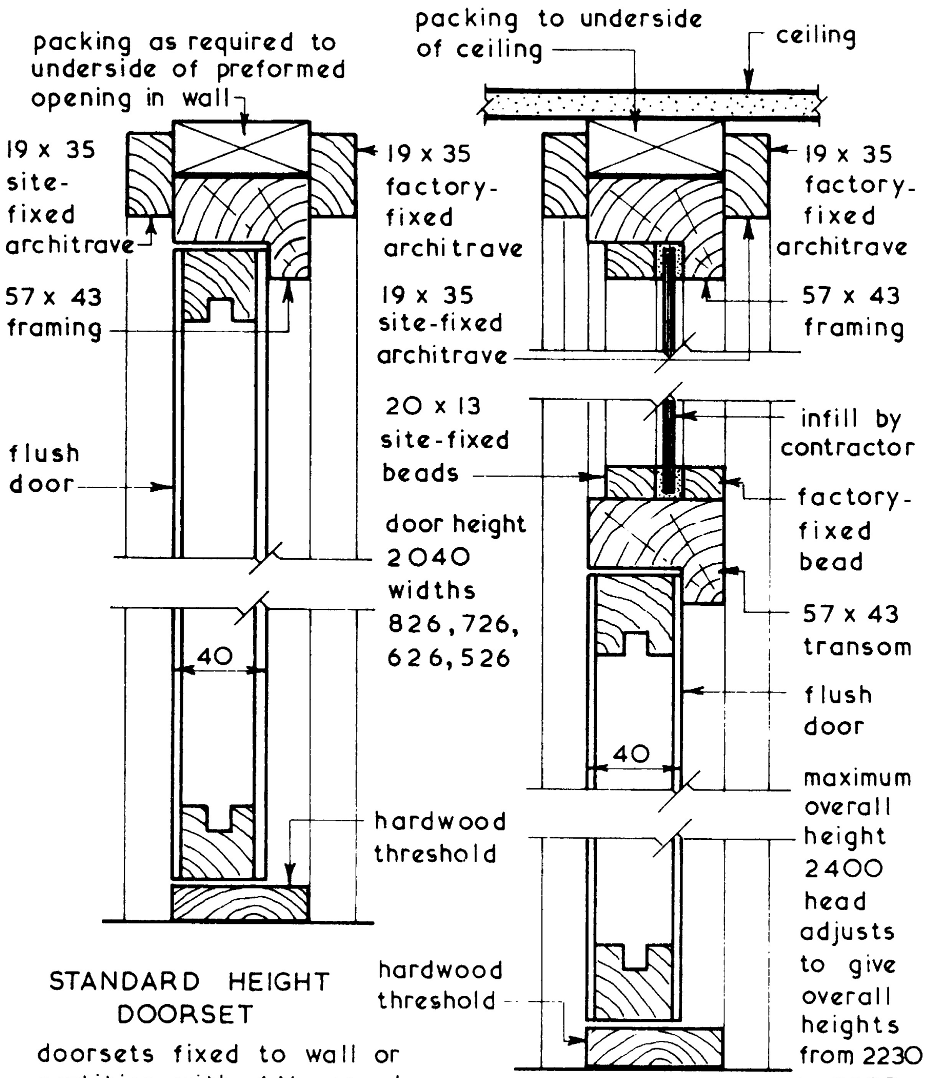
STANDARD FRAME

STOREY HEIGHT FRAME

Ref. BS 4787: Internal and external wood doorsets, door leaves and frames. Specification for dimensional requirements.

Doorsets ~ these are factory-produced fully assembled prehung doors which are supplied complete with frame, architraves and hardware except for door furniture. The doors may be hung to the frames using pin butts for easy door removal. Prehung door sets are available in standard and storey height versions and are suitable for all internal door applications with normal wall and partition thicknesses.

Typical Examples ~



STANDARD HEIGHT DOORSET

doorsets fixed to wall or partition with 4 No. wood screws to each jamb

STOREY HEIGHT DOORSET

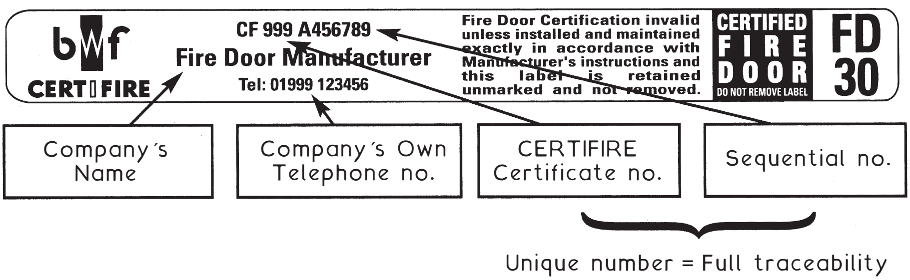
Fire Doorset ~ a 'complete unit consisting of a door frame and a door leaf or leaves, supplied with all essential parts from a single source'. The difference between a doorset and a fire doorset is that the latter is endorsed with a fire certificate for the complete unit. When supplied as a collection of parts for site assembly, this is known as a door kit.

Fire Door Assembly ~ a 'complete assembly as installed, including door frame and one or more leaves, together with its essential hardware [ironmongery] supplied from separate sources'. Provided the components to an assembly satisfy the Building Regulations – Approved Document B, fire safety requirements and standards for certification and compatibility, then a fire door assembly is an acceptable alternative to a doorset.

Fire doorsets are usually more expensive than fire door assemblies, but assemblies permit more flexibility in choice of components. Site fixing time will be longer for assemblies.

(Quotes from BS EN 12519: Windows and pedestrian doors. Terminology.)

Fire Door ~ a fire door is not just the door leaf. A fire door includes the frame, ironmongery, glazing, intumescent core and smoke seal. To comply with European market requirements, ironmongery should be CE marked (see page 79). A fire door should also be marked accordingly on the top or hinge side. The label type shown below, reproduced with kind permission of the British Woodworking Federation, is acceptable.



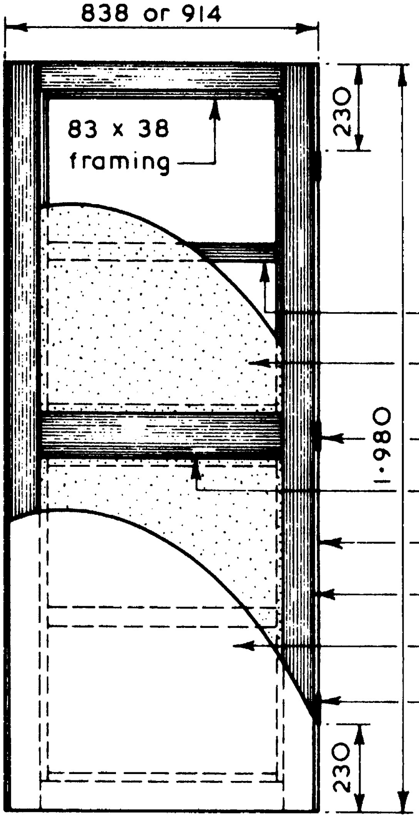
Ref. BS 8214: Code of practice for fire door assemblies.

30-minute Flush Fire Doors ~ these are usually based on the recommendations given in BS 8214. A wide variety of door constructions are available from various manufacturers but they all have to be fitted to a similar frame for testing as a doorset or assembly, including ironmongery.

A door's resistance to fire is measured by:

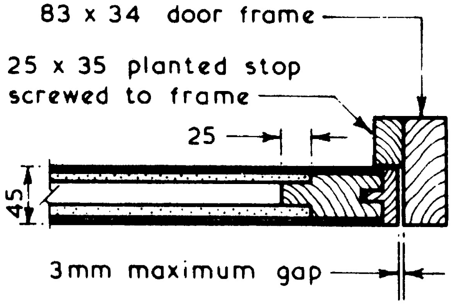
1. Insulation – resistance to thermal transmittance, see BS 476-20 & 22: Fire tests on building materials and structures.
2. Integrity – resistance in minutes to the penetration of flame and hot gases under simulated fire conditions.

Typical Details ~



SECTIONAL ELEVATION

NB. intumescent strips in door edges provide an enhanced rating – see page 885.

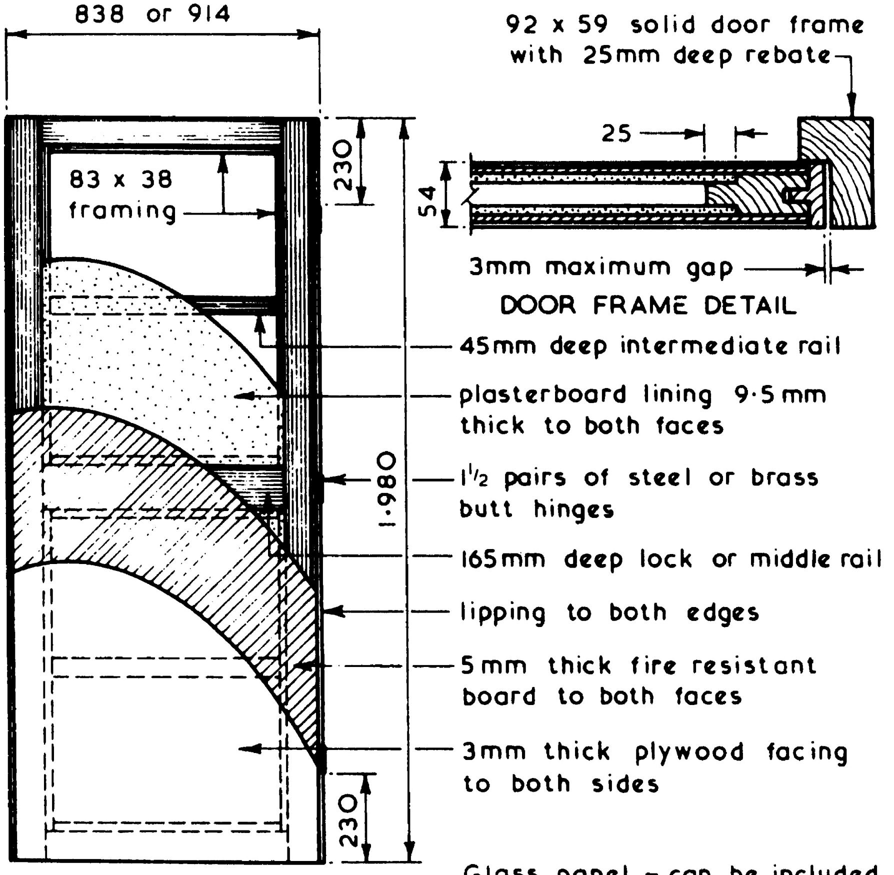


DOOR FRAME DETAIL

Glass panel – can be included if 6mm thick wired glass with a maximum size of $1 \cdot 2\ m^2$ is used. Glass to be fixed with non-combustible beads with a melting point of ≮ 900°C. See also page 886.

60-minute Flush Fire Doors ~ like the 30-minute flush fire door shown on the previous page these doors are based on the recommendations given in BS 8214 which covers both door and frame. A wide variety of fire resistant door constructions are available from various manufacturers with most classified as having both insulation and integrity ratings of 60 minutes.

Typical Details ~



SECTIONAL ELEVATION

NB. intumescent strips in door edges and frame rebate would give above door an enhanced rating ~ see next page.

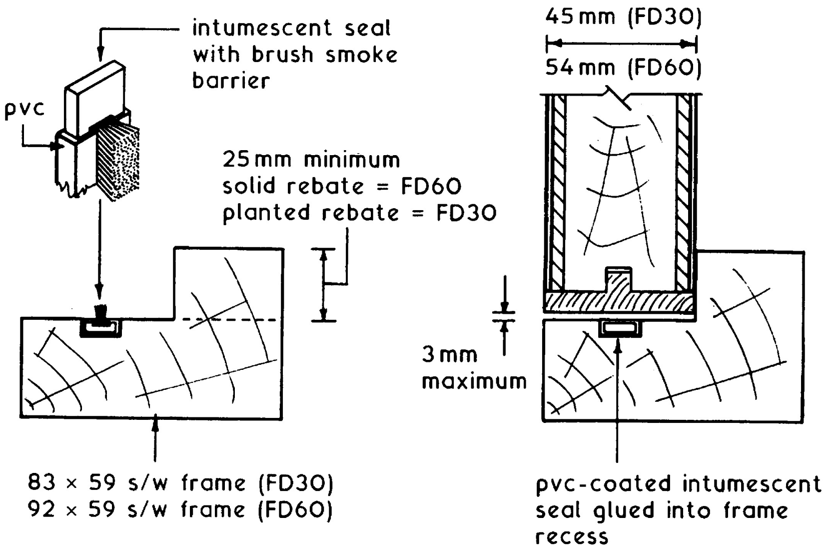
Glass panel – can be included if 6mm thick wired glass with a maximum size of $0{\cdot}5\ m^2$ is used. Glass to be fixed with non-combustible beads with a melting point of $\nless 900°C$.
See also page 886.

Ref. BS 8214: Code of practice for fire door assemblies.

Fire and Smoke Resistance ~ doors can be assessed for both integrity and smoke resistance. They are coded accordingly, for example, FD30 or FD30s. FD indicates a fire door and 30 the integrity time in minutes. The letter 's' denotes that the door or frame contains a facility to resist the passage of smoke.

Manufacturers produce doors of standard ratings – 30, 60 and 90 minutes, with higher ratings available to order. A colour-coded plug inserted in the door edge corresponds to the fire rating. See BS 8214, Table 1 for details.

Intumescent Fire and Smoke Seals ~



The intumescent core may be fitted to the door edge or the frame. In practice, most joinery manufacturers leave a recess in the frame where the seal is secured with rubber-based or PVA adhesive. At temperatures of about 150°C, the core expands to create a seal around the door edge. This remains throughout the fire-resistance period whilst the door can still be opened for escape and access purposes. The smoke seal will also function as an effective draught seal.

Further refs.:

BS EN 1634-1: Fire resistance and smoke control tests for door, shutter and openable window assemblies and elements of building hardware.

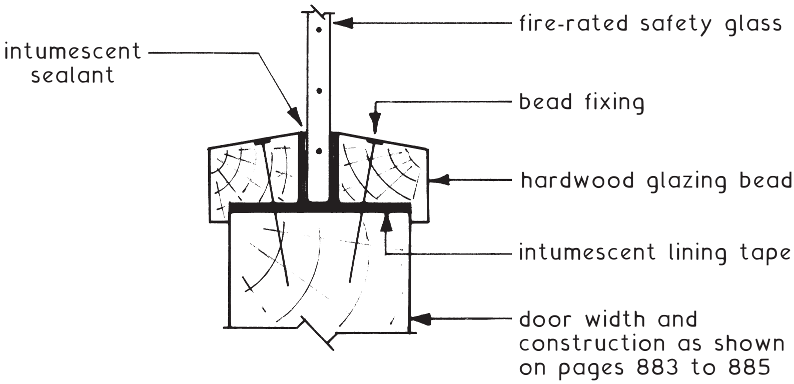
BS EN 13501: Fire classification of construction products and building elements.

Apertures will reduce the potential fire resistance if not appropriately filled. Suitable material should have the same standard of fire performance as the door into which it is fitted.

Fire-rated glass types:

- Embedded Georgian wired glass
- Composite glass containing borosilicates and ceramics
- Tempered and toughened glass
- Glass laminated with reactive fire-resisting interlayers

Installation ~ hardwood beads and intumescent seals. Compatibility of glass type and sealing product is essential, therefore manufacturer's details must be consulted.



Intumescent products:

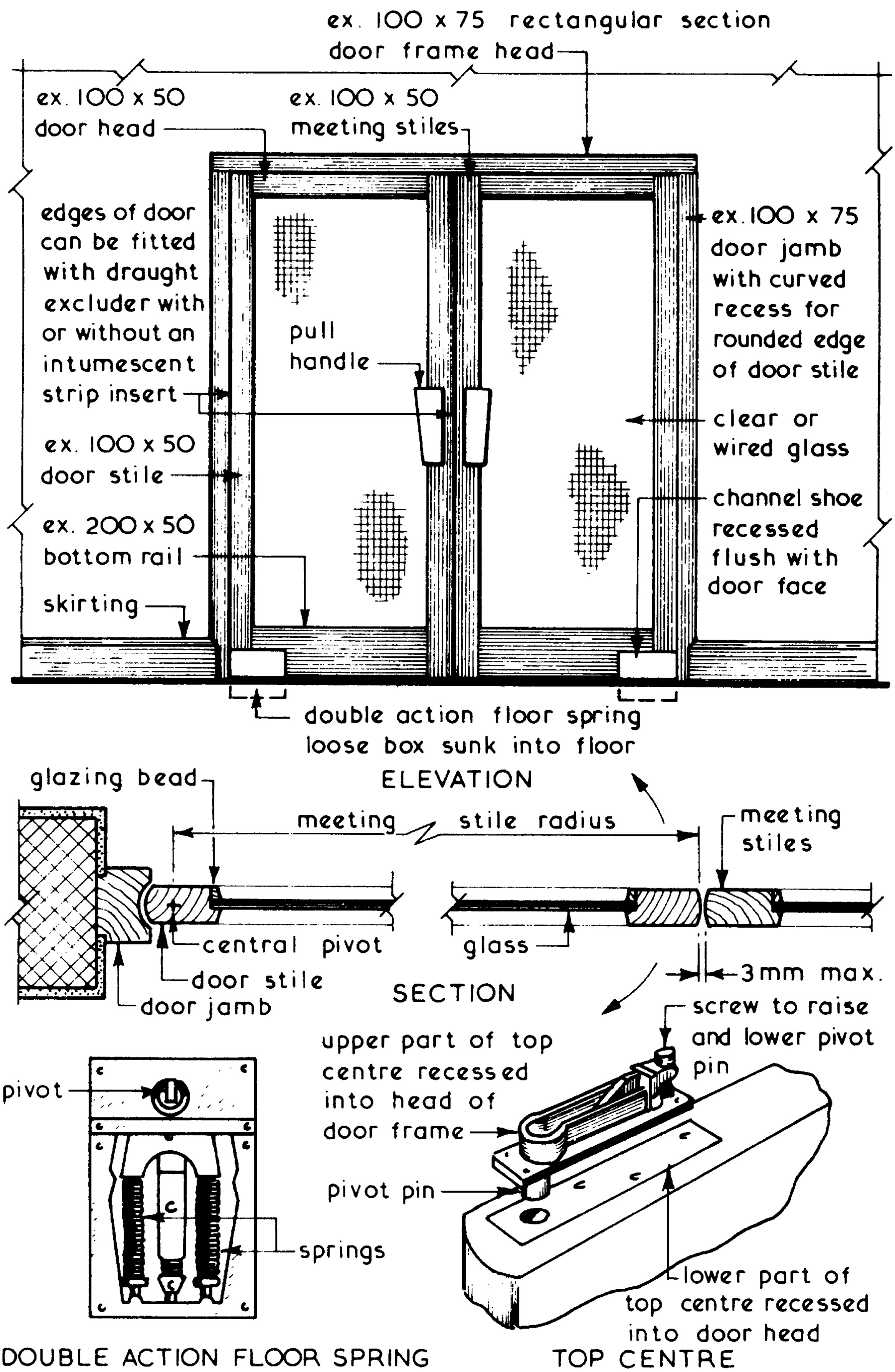
- Sealants and mastics 'gun' applied
- Adhesive glazing strip or tape
- Preformed moulded channel

Note: Calcium silicate preformed channel is also available. Woven ceramic fire-glazing tape/ribbon is produced specifically for use in metal frames.

Building Regulations refs.:

A.D. M, Section 3.10, visibility zones/panels between 500 and 1500 mm above floor finish, and A.D. K, Section 10.
A.D. K, Section 5, aperture size (see page 472).

Typical Details ~



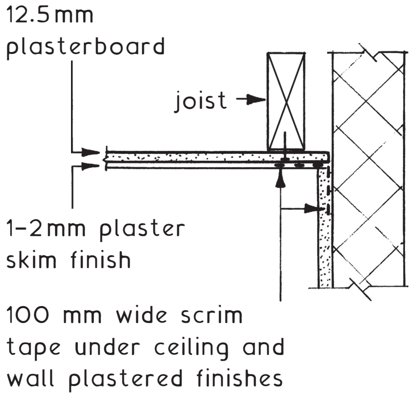
Plasterboard ~ a rigid board composed of gypsum sandwiched between durable lining paper outer facings. For ceiling applications, the following types can be used:

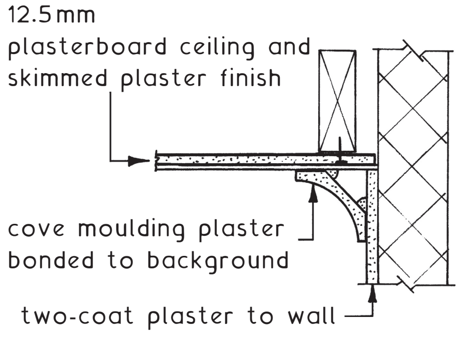
Baseboard -
1220 × 900 × 9.5 mm thick for joist centres up to 400 mm.
1220 × 600 × 12.5 mm thick for joist centres up to 600 mm.

Baseboard has square edges and can be plaster skim finished. Joints are reinforced with self-adhesive 50 mm min. width glass fibre mesh scrim tape or the board manufacturer's recommended paper tape. These boards are also made with a metallised polyester foil backing for vapour check applications. The foil is to prevent any moisture produced in potentially damp situations such as a bathroom or in warm roof construction from affecting loft insulation and timber. Joints should be sealed with an adhesive metallised tape.

Wallboard –
9.5, 12.5 and 15 mm thicknesses, 900 and 1200 mm widths and lengths of 1800 and 2400 mm. Longer boards are produced in the two greater thicknesses and a vapour check variation is available. Edges are either tapered for taped and filled joints for dry lining or square for skimmed plaster or textured finishes.

Plasterboards should be fixed breaking joint to the underside of floor or ceiling joists with zinc plated (galvanised) nails or dry-wall screws at 150 mm max. spacing. The junction at ceiling to wall is reinforced with glass fibre mesh scrim tape or a preformed plaster moulding.



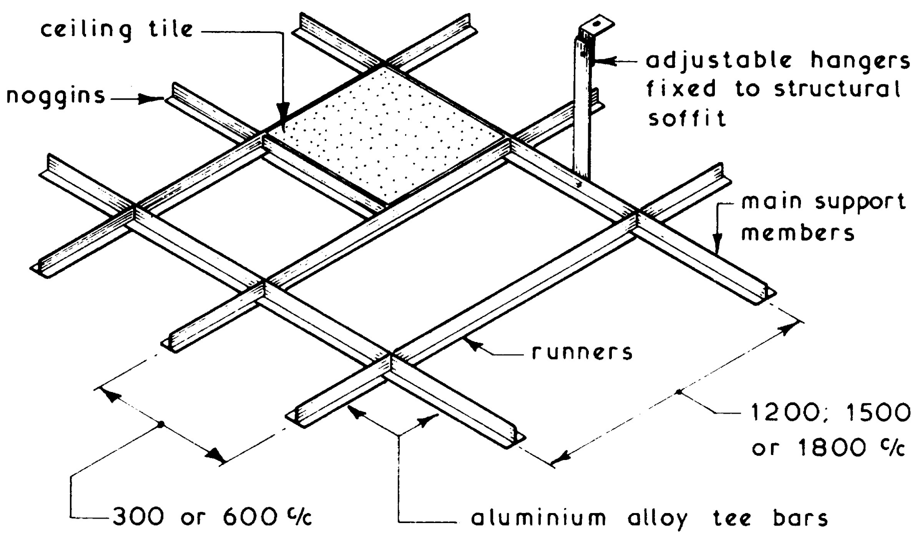


Alternative

Suspended Ceilings ~ these can be defined as ceilings which are fixed to a framework suspended from main structure, thus forming a void between the two components. The basic functional requirements of suspended ceilings are:

1. They should be easy to construct, repair, maintain and clean.
2. So designed that an adequate means of access is provided to the void space for the maintenance of the suspension system, concealed services and/or light fittings.
3. Provide any required sound and/or thermal insulation.
4. Provide any required acoustic control in terms of absorption and reverberation.
5. Provide if required structural fire protection to structural steel beams supporting a concrete floor and contain fire stop cavity barriers within the void at defined intervals.
6. Conform with the minimum requirements set out in the Building Regulations governing the restriction of spread of flame over surfaces of ceilings and the exemptions permitting the use of certain plastic materials.
7. Flexural design strength in varying humidity and temperature.
8. Resistance to impact.
9. Designed on a planning module, preferably a 300 mm dimensional coordinated system.

Typical Suspended Ceiling Grid Framework Layout ~

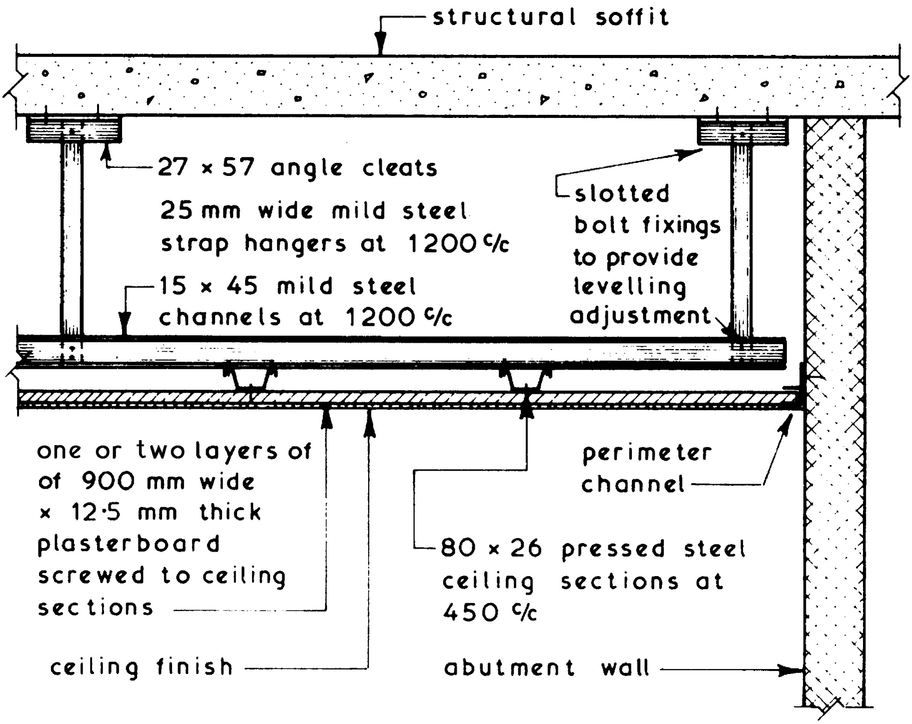


Classification of Suspended Ceiling ~ there is no standard method of classification since some are classified by their function such as illuminated and acoustic suspended ceilings, others are classified by the materials used and classification by method of construction is also very popular. The latter method is simple, since most suspended ceiling types can be placed in one of three groups:

1. Jointless suspended ceilings.
2. Panelled suspended ceilings – see next page.
3. Decorative and open suspended ceilings – see page 892.

Jointless Suspended Ceilings ~ these forms of suspended ceilings provide a continuous and jointless surface with the internal appearance of a conventional ceiling. They may be selected to fulfil fire resistance requirements or to provide a robust form of suspended ceiling. The two common ways of construction are a plasterboard or expanded metal lathing soffit with hand-applied plaster finish or a sprayed applied rendering with a cement base.

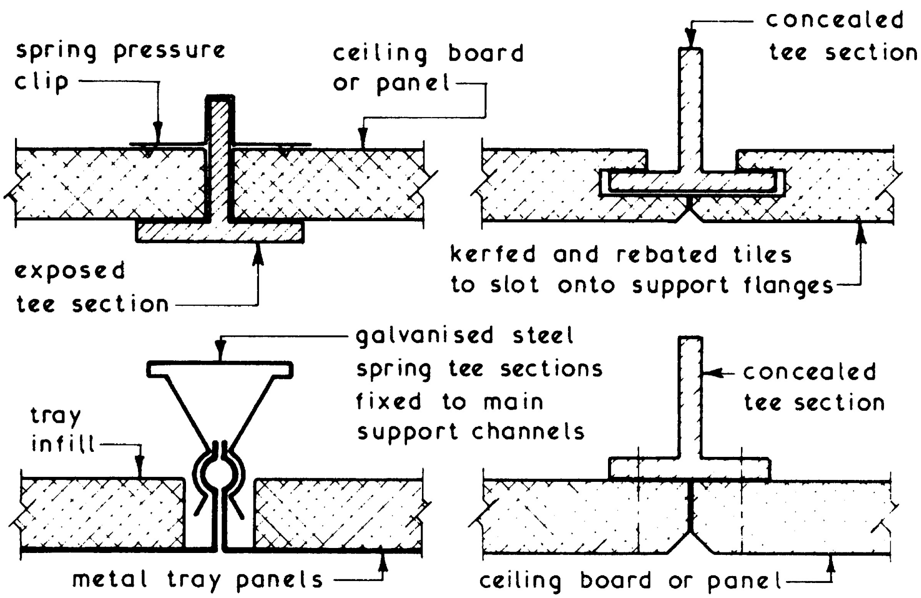
Typical Details ~



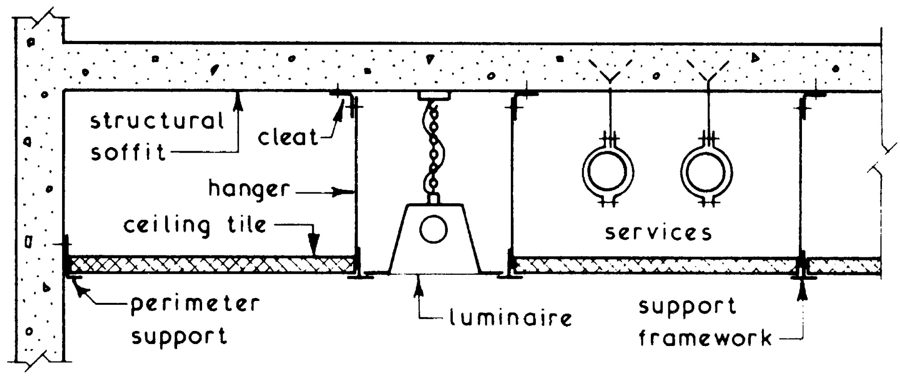
See also: BS EN 13964: Suspended ceilings. Requirements and test methods.

Panelled Suspended Ceilings ~ these are the most popular form of suspended ceiling consisting of a suspended grid framework to which the ceiling covering is attached. The covering can be of a tile, tray, board or strip format in a wide variety of materials with an exposed or concealed supporting framework. Services such as luminaries can usually be incorporated within the system. Generally panelled systems are easy to assemble and install using a water level or laser beam for initial and final levelling. Provision for maintenance access can be easily incorporated into most systems and layouts.

Typical Support Details ~

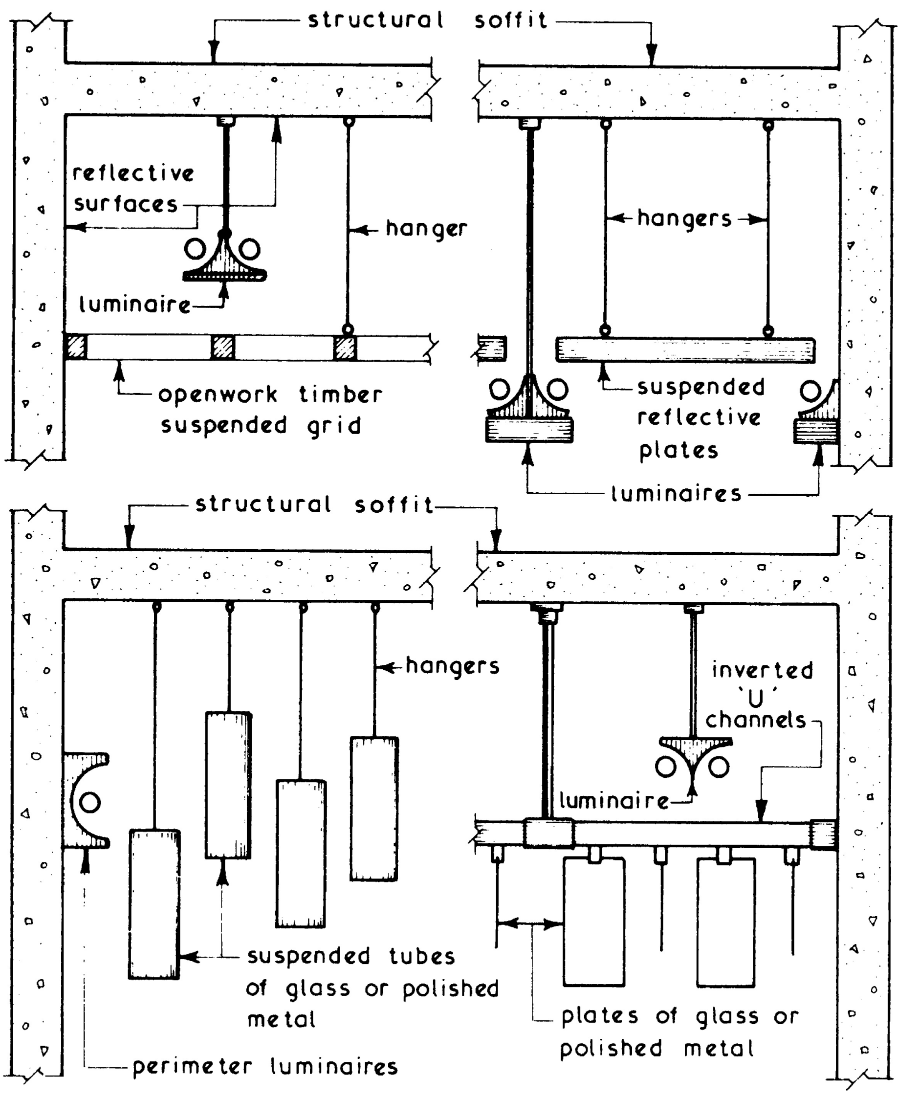


Typical Panelled Suspended Ceiling Details ~

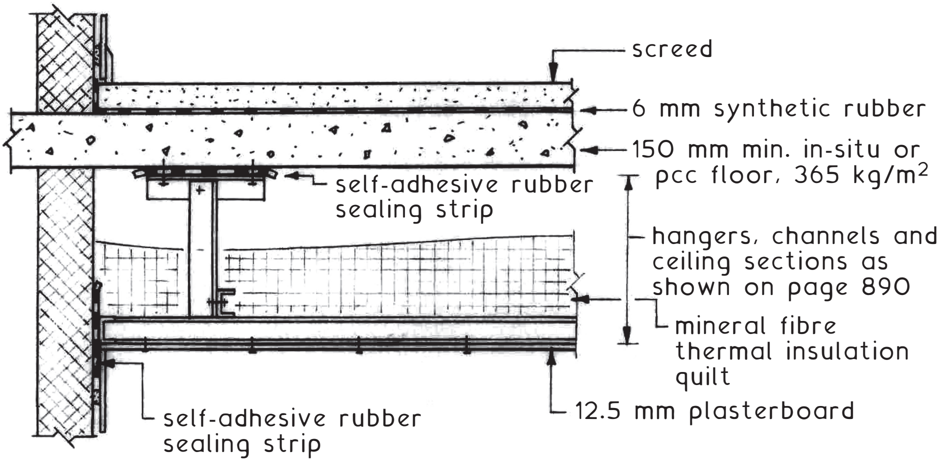


Decorative and Open Suspended Ceilings ~ these ceilings usually consist of an openwork grid or suspended shapes onto which the lights fixed at, above or below ceiling level can be trained, thus creating a decorative and illuminated effect. Many of these ceilings are purpose designed and built as opposed to the proprietary systems associated with jointless and panelled suspended ceilings.

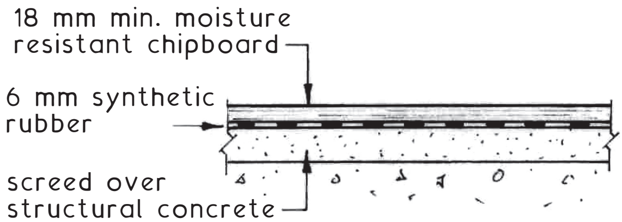
Typical Examples ~



Separating floor construction for sound insulation purposes. References: Robust Details, see page 71 and Building Regulations, Approved Document E: Resistance to the passage of sound.



Alternative Finish with Floating Deck ~



Further Building Regulations References ~

Approved Document B: Fire safety, Volumes 1 and 2 – Fire resisting cavity barriers in concealed spaces. Located at compartmented divisions within a building (see page 18), otherwise at a maximum spacing that will depend on the situation, building type and its function. Minimum fire resistance is 30 minutes but often more depending on the building use and purpose.

Approved Document L: Conservation of fuel and power – thermal insulation requirements.